INTERNATIONAL JOURNAL OF PROGRESSIVE RESEARCH IN ENGINEERING MANAGEMENT AND SCIENCE (IJPREMS)
(Int Peer Reviewed Journal)
Vol. 06, Issue 03, March 2026, pp : 843-852
www.ijprems.com
editor@ijprems.com
e-ISSN : 2583-1062
Impact Factor : 7.001

FACTORY-TO-CUSTOMER SHIPPING ROUTE EFFICIENCY ANALYSIS USING THE NASSAU CANDY DISTRIBUTOR DATASET

Akshay Tangade[1], Saiprasad Kagne[2]
[1]Business Analyst, Unified Mentor, Pune, India.
[2]AI Powered Data Scientist, Unified Mentor, Delhi, India
DOI: https://www.doi.org/10.58257/IJPREMS51848

ABSTRACT

This paper shows the analytical methodology, implementations, design and visual findings of a shipping route efficiency intelligence system developed for Nassau Candy Distributor Factory. The system works on 10,194 order records spanning from 2024 to 2025, and carries a structured analytical procedure to clean, enrich, and aggregate the data. It calculates five Key Performance Indicators (KPIs) and presents the results in an interactive five-tab Streamlit dashboard. There is an identified systematic data anomaly in the years of the ship dates, which has been corrected by using modulo 365. From the data analysis, 196 unique factory-to-state shipping routes, six geographic bottleneck states, and the performance difference between four modes of shipping were identified. Dashboard screenshots are included throughout the paper to show how the analytical outputs are presented to end users.

Keywords: Shipping Route Efficiency, Lead Time Analysis, KPI Computation, Geographic Bottleneck Detection, Data Anomaly Correction, Logistics Dashboard, Streamlit, Supply Chain Optimisation.

1. INTRODUCTION

I.1 Background and Business Context

Nassau Candy Distributor is a company in speciality confections and fine foods distribution. It supplies products to customers across the United States and Canada. The company operates five production factories, each manufacturing a distinct portfolio of products, and dispatches orders through four shipping modes: Same Day, First Class, Second Class, and Standard Class.

Even after processing thousands of orders annually, the organisation was not able to create a systematic mechanism that can evaluate the efficiency of individual shipping routes, compare the cost-speed trade-offs of different shipping methods, or identify geographic areas of poor performance.

I.2 Problem Statement

There were three core challenges. First, the raw transactional data contained a critical quality issue, ship date years were systematically inflated, making all lead time calculations meaningless without correction. Second, no analytical framework existed to transform raw order data into actionable route-level intelligence. Third, the findings needed to be accessible to non-technical stakeholders through an interactive visual interface.

I.3 Project Objectives

- Detect and resolve data quality issues in the source dataset
- Implement a complete analytical methodology pipeline
- Compute all five KPIs defined in the BRD for every shipping route
- Identify the most and least efficient routes, geographic bottlenecks, and ship mode differentials
- Deliver findings through a production-quality interactive Streamlit dashboard

I.4 Scope

The analysis covers order data from 2nd January 2024 to 31st December 2025, encompassing 10,194 order records, 15 product SKUs, 5 factories, 4 shipping modes, and deliveries to 59 states and provinces across four geographic regions.

2. DATASET DESCRIPTION

II.1 Source Data

Nassau Candy Distributor.csv is the source dataset which contains 10,194 rows and 18 columns. Each row represents one product ordered on one date, shipped to one customer location.

Table 1: Dataset Column Reference

Column	Description
Order Date	Date the customer placed the order (DD-MM-YYYY)
Ship Mode	Same Day/ First Class/ Second Class/ Standard Class
Ship Date	Date the order was shipped – contains year inflation anomaly
State/Province	Customer delivery state or Canadian province
Product Name	Full product name (15 unique SKUs)
Region	Broad geographic region: Interior, Atlantic, Gulf, Pacific
Sales	Revenue generated in USD
Gross Profit	Profit in USD after cost deduction
Units	Quantity of units ordered
Cost	Unit cost in USD

II.2 Factory and Product Structure

Table 2: Product-to-Factory Mapping

Factory	Products Manufactured	Order Volume
Lot's O' Nuts	Wonka Bar (Nutty Crunch, Fudge Mallows, Scrumdiddlyumptious)	5,692
Wicked Choccy's	Wonka Bar(Milk Chocolate, Triple Dazzle Caramel)	4,152
Secret Factory	Everlasting Gobstopper, Lickable Wallpaper, Wonka Gum	217
The Other Factory	Hair Toffee, Kazookles	100
Sugar Shack	Laffy Taffy, SweeTARTS, Nerds, Fun Dip, Fizzy Lifting Drinks	33

3. DATA QUALITY AND ANOMALY DETECTION

III.1 Completeness Assessment

Carrying out a systematic check for null values in all 18 columns for the 10,194 rows, found zero instances of missing values. The financial columns, namely Sales, Units, Gross Profit, and Cost, were also checked for negative values. All four columns reported zero instances of negative values.

III.2 The Ship Date Anomaly

When the lead time was computed directly from the data using the formula (Ship Date – Order Date), the values ranged from 904 to 1,642 days or 2.5 to 4.5 years. The data was found to have three distinct groupings with a difference of exactly 365 days. This indicates that the year for the Ship Date was incorrectly entered by 2, 3, or 4 years in the original data.

III.3 Anomaly Resolution

The resolution to this anomaly was to use the following formula to correct the data: actual_lead_time = raw_lead_time % 365. The reason for this calculation lies in the fact that all three groupings have the same remainder when divided by 365. Therefore, dividing by 365 would yield the actual shipping time. The range of the corrected data was 174 to 185 days, a reasonable time for shipping within the US.

Table 3: Lead Time Before and After Anomaly Correction

Statistic	Raw Lead Time (days)	Corrected Lead Time (days)
Minimum	904	174
Mean	1,320.8	178.3
Median	1,274	178
Maximum	1,642	185
Std Deviation	262.4	1.8
Interpretation	Impossible (2-4 years)	Realistic (6 months)

4. ANALYTICAL METHODOLOGY

The analytical methodology is implemented step-by-step in a Python script named analysis.py. All computation is strictly separate from presentation, and eleven CSV output files are produced, which the dashboard reads without requiring recomputation.

IV.1 Loading, Cleaning, Feature Engineering

The data loading and feature engineering steps include loading the raw data from a CSV file into a Pandas DataFrame, applying a modulo 365 date correction, stripping whitespace from geographic data, and creating six new features: Factory (from product to factory), Route_State (Factory to State), Route_Region (Factory to Region), Order Year, Order Month, and Order Quarter.

IV.2 KPI Computation

Table 4: KPI Definitions and Formulas

KPI	Definition and Formula
KPI 1: Shipping Lead Time	(Ship Date - Order Date).dt.days % 365 per order
KPI 2: Average Lead Time	Mean Lead Time per route group
KPI 3: Route Volume	Count of orders per route group
KPI 4: Delay Frequency	% of orders exceeding the 75th percentile threshold (179 days)
KPI 5: Route Efficiency Score	100 × (1 - (Lead Time - min) / (max - min)) {normalised 0 to 100}

IV.3 Aggregation and Benchmarking

The next steps aggregate 10,194 orders in to 196 route summaries using groupby aggregation. Later, it calculates top and bottom 10 benchmarks and aggregates again by ship mode. After that, it calculates dual-threshold bottleneck detection based on P60 for both volume and lead time. And finally ship mode KPI's are calculated, including a pivot table for Factory by Ship Mode. Step 9 calculates factory summary data.

5. RESULTS AND KEY FINDINGS

V.1 Overall Performance Summary

Table 5: Global Performance KPIs

KPI	Value
Total Shipments Analysed	10,194
Date Range	02 Jan 2024 – 31 Dec 2025
Unique Factory-to-State Routes	196
Overall Average Lead Time	178.3 days
Overall Delay Rate (above P75)	24.7%
Delay Threshold (P75)	179 days
Lead Time Range	174 – 185 days (11-day spread)
Average Efficiency Score	61.0 / 100

V.2 Route Efficiency Benchmarks

Table 6: Top 5 and Bottom 5 Routes by Efficiency

Route	Avg Lead Time	Delay Rate	Eff. Score
The Other Factory → North Carolina	175.0 days	0.0%	90.9
The Other Factory → Kentucky	176.0 days	0.0%	81.8
Secret Factory → New Mexico	176.0 days	0.0%	81.8
Sugar Shack → Connecticut	181.0 days	100.0%	36.4
Lot's O' Nuts → Newfoundland & Labrador	182.0 days	83.3%	27.3
Sugar Shack → New Jersey	182.0 days	100.0%	27.3

V.3 Identified Bottleneck States

Table 7: Six Geographic Bottleneck States

State	Shipments	Avg Lead Time	Delay Rate
Minnesota (Interior)	89	178.96 days	31.5%
Tennessee (Gulf)	183	178.81 days	35.0%
New Jersey (Atlantic)	130	178.75 days	35.4%
Indiana (Interior)	149	178.71 days	34.2%
Oregon (Pacific)	124	178.60 days	33.9%

State	Shipments	Avg Lead Time	Delay Rate
Delaware (Atlantic)	96	178.55 days	32.3%

V.4 Ship Mode Performance

Table 8: Ship Mode Performance Comparison

Ship Mode	Avg Lead Time	Delay Rate	Avg Unit Cost
Same Day	174.4 days	0.0%	$4.41
First Class	176.6 days	0.0%	$4.72
Second Class	177.6 days	7.5%	$4.83
Standard Class	179.3 days	38.8%	$4.75

6. SYSTEM ARCHITECTURE

VI.1 Two-Layer Design

The system is designed in such a way that it clearly separate the computation(analysis.py) from the presentation (dashboard.py). In the first layer, the code reads the data, calculates all the KPIs, and generates eleven CSV files. In the second layer, the code simply reads the eleven CSVs and presents them visually.

VI.2 Output File Structure

Table 9: Eleven Output Files Produced by analysis.py

Output File	Contents
kpi_orders_enriched.csv	All 10,194 orders with Lead Time, Factory, Route, Is_Delayed, Efficiency_Score
kpi_route_aggregation.csv	196 routes with all 12 aggregated KPI metrics — dashboard backbone
benchmark_top10_routes.csv	Top 10 fastest routes
benchmark_bottom10_routes.csv	Bottom 10 slowest routes
kpi_routes_by_ship_mode.csv	506 route records across all 4 ship modes
kpi_state_performance.csv	59 states with KPIs and bottleneck flag
kpi_region_performance.csv	4 regional KPI summaries
kpi_bottleneck_states.csv	6 identified bottleneck states
kpi_ship_mode_performance.csv	4 ship mode KPI profiles
kpi_factory_shipmode_pivot.csv	5×4 Factory × Ship Mode lead time matrix
kpi_factory_performance.csv	5 factory KPI summaries

VI.3 Technology Stack

- Python 3.12 - core language
- Pandas 2.x - data transformation, groupby aggregation, CSV I/O
- NumPy - numerical operations for KPI calculations
- Streamlit - web application framework

- Plotly Express / Graph Objects - interactive charts (choropleth, box, scatter, heatmap, bar)
- Pillow (PIL) - logo image rendering in the dashboard

7. DASHBOARD DESIGN AND FUNCTIONALITY

VII.1 Design Philosophy

The dashboard was built such that every question that a logistics manager would want to ask about shipping performance is answerable with no more than two clicks. The dark theme KPI cards use blue as an accent colour and gold borders similar to the Nassau Candy branding. Red is used consistently to represent poor performance and green to represent good performance across all charts.

The @st.cache_data decorator loads the eleven CSV files into memory once and then uses them across all subsequent user interactions, keeping the app fast and responsive no matter how many filters or tabs the user uses.

VII.2 Tab 1: Route Efficiency Overview

The first tab offers a comprehensive view of all 196 routes. Five KPI summary cards provide totals such as 10,194 total shipments, an average lead time of 178.3 days, an average delay rate of 24.0%, unique routes totalling 196, and an average efficiency score of 60.6/100. The Bottom 10 routes and the Top 10 routes are shown in two horizontal bar charts below the summary cards. The routes are coloured according to their efficiency status: red, yellow, and green.

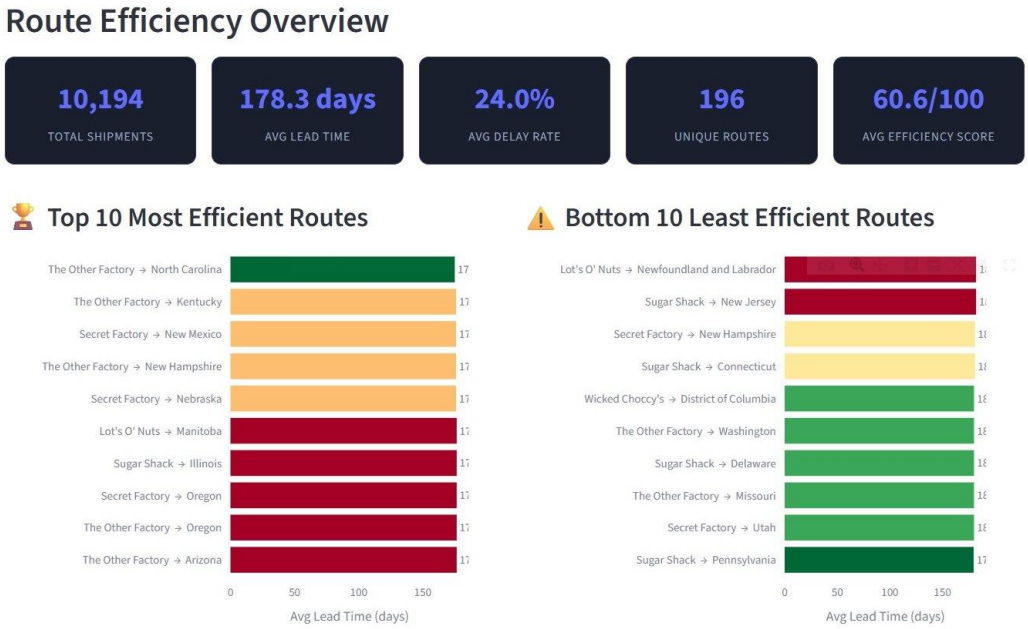

Figure 1: Tab 1: Route Efficiency Overview showing KPI cards and Top 10 / Bottom 10 bar charts

Below the benchmarks, the scatter plot shows every route in terms of the total shipment volume and the efficiency score. The size of the bubble represents the delay rate, and the color represents the delay intensity, ranging from green for less delay to red for high delay. This chart shows that high-volume routes don't necessarily mean that they're better managed because several high-volume routes fall in the middle of the efficiency score.

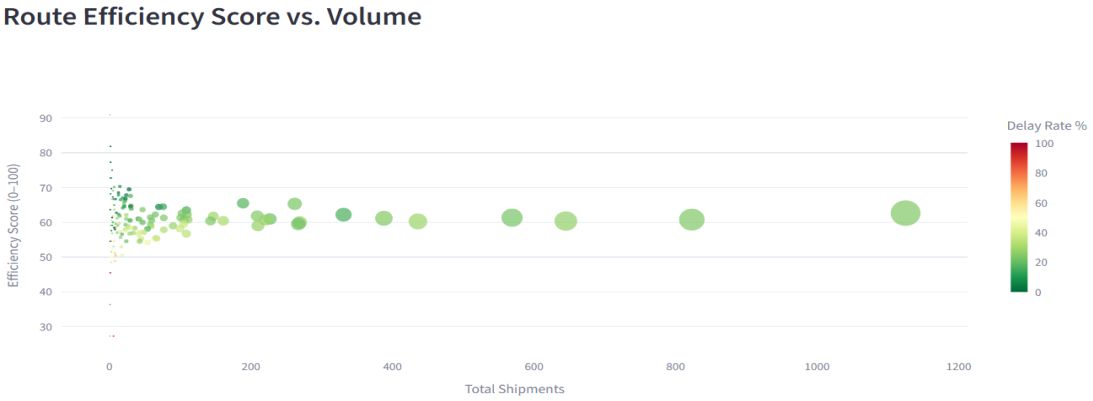

Figure 2: Tab 1: Route Efficiency Score vs. Total Shipments scatter plot (bubble size = delay rate, colour = delay intensity)

VII.3 Tab 2: Geographic Shipping Performance Map

The geographic tab displays the US map in which the states are coloured based on the selected data. Three radio buttons allow the user to choose the data that will be used for the map. Options include Average Lead Time, Delay Rate, and Total Shipments. Overlaid on the map are the factory locations, represented by yellow stars that show the location of the factories.

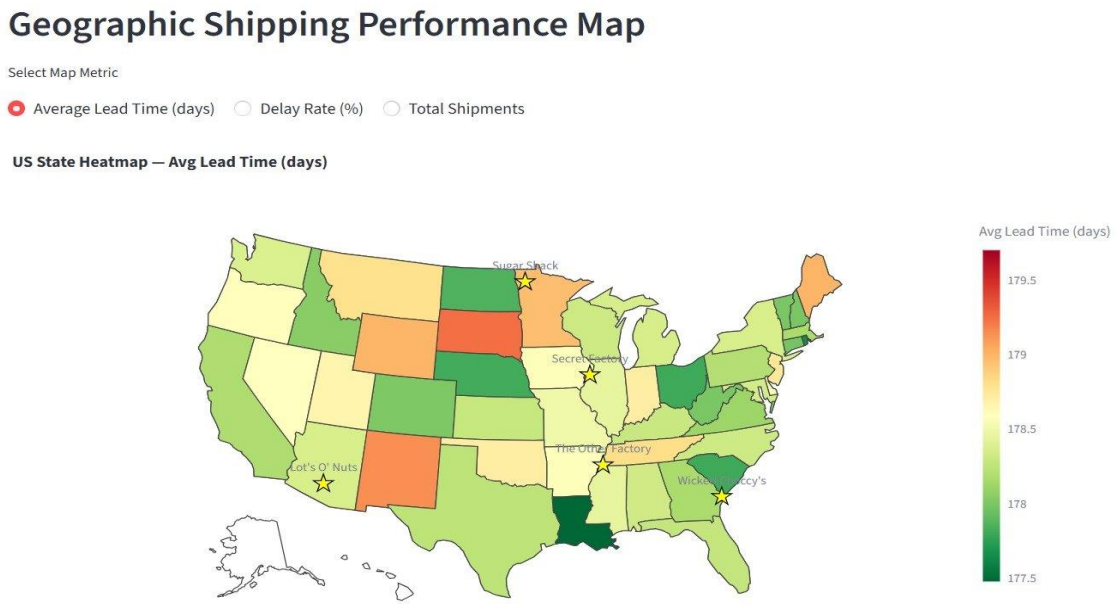

Figure 3: Tab 2: Geographic Shipping Performance Map showing average lead time by US state with factory locations marked as yellow stars

The map in Figure 3 illustrates the range of lead time, from deepest green for the fastest (Louisiana, Ontario, Ohio), through shades of green and yellow, to orange-red for the slowest (Newfoundland, South Dakota, New Mexico). The five factory star markers give context for understanding the reason for differential service speeds for some states relative to their proximity to factory locations

VII.4 Tab 3: Geographic Bottleneck Analysis

This tab uses the BRD bottleneck detection requirement as an analytical view. The centrepiece is a quadrant chart that plots each state on the chart using shipment volume on the X-axis and average lead time on the Y-axis. Two dashed lines are drawn at the 60th percentile thresholds for each axis, dividing the chart into four sections. States that fall into the top-right section, those that are both high in shipment volume and high in average lead time, are highlighted in red as the operational bottlenecks.

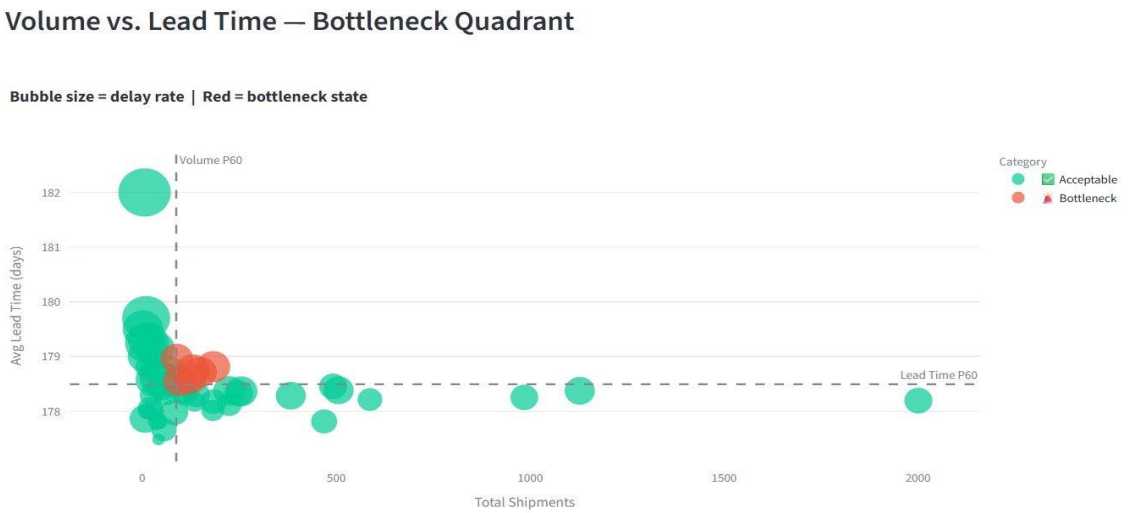

Figure 4: Tab 3: Volume vs. Lead Time Bottleneck Quadrant. Red bubbles in the top-right quadrant are identified bottleneck states. Bubble size represents delay rate.

As can be seen from Figure 4, the six bottleneck states are concentrated in the top-right section of the chart, falling between the range of 89 and 183 in shipment volume, and between 178.55 and 178.96 average lead time days. All six states have delay rates between 31 and 35%, the highest range of any state.

7.5 Tab 4: Ship Mode Performance Analysis

The ship mode tab satisfies the requirement of the BRD to compare Standard vs. Expedited shipping by providing a cost-time descriptive analysis. Four metric cards at the top of the page immediately present the key result: Same Day and First Class both have 0.0% delay rates, Second Class has 7.5%, while Standard Class has 38.8%, meaning nearly 4 out of every 10 Standard Class packages had delays.

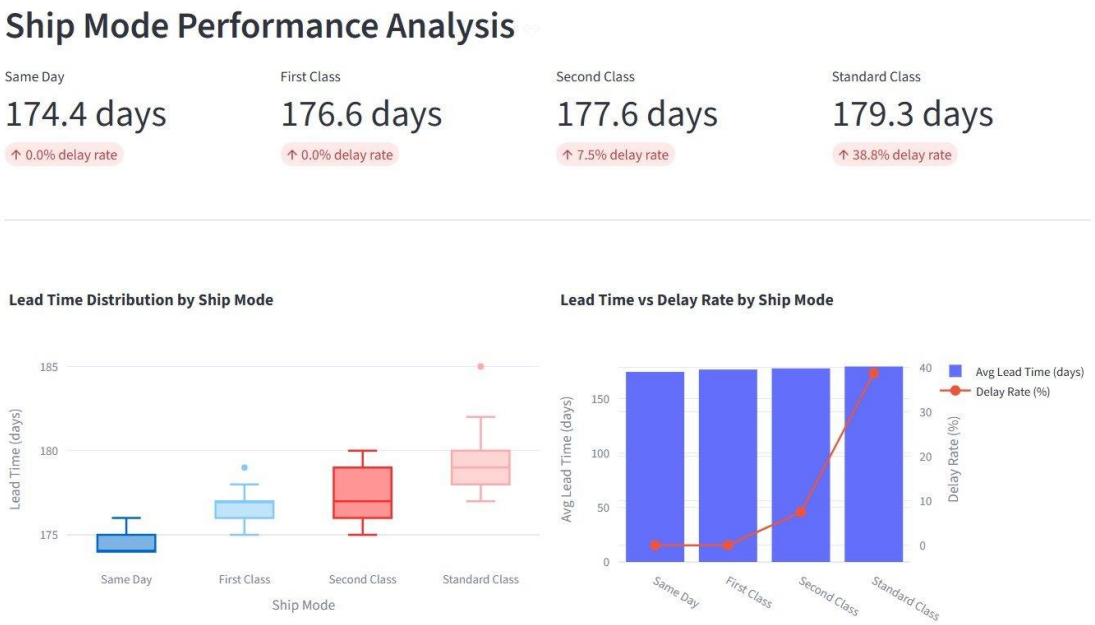

Figure 5: Tab 4: Ship Mode Performance Analysis showing KPI metric cards, lead time distribution box plots, and the lead time vs. delay rate dual-axis chart

The box plot on the left of Figure 5 shows that Same Day has the lowest spread, while Standard Class has the largest box spread and the only value above 185 days. The dual-axis chart on the right shows average lead time (bars) overlaid with delay rate (red line), allowing the reader to easily see that the delay rate jumps sharply from Second Class to Standard Class even though the lead time difference is not large, less than 2 days.

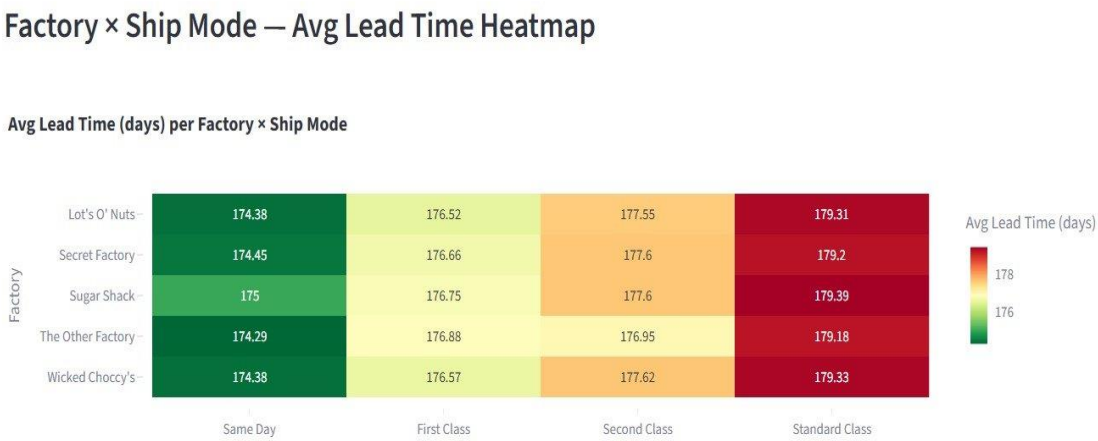

Figure 6: Tab 4: Factory × Ship Mode Average Lead Time Heatmap. Deep green = fastest, deep red = slowest.

As seen in Figure 6, there is a clear trend in all five factories. Same Day (174.3-175.0 days) is always green, and Standard Class (179.2-179.4 days) is always red. Other Factory consistently has the lowest values in all modes, reinforcing its position as the best factory in the network. Most importantly, there is no factory that stands out as better or worse in any given mode. The mode is dominant, not the factory.

VII.6 Tab 5: Route Drill-Down

Finally, there is a tab that offers detailed information about any given route in the Factory-State space. Two dropdown menus allow the user to select a factory and then a destination state, filtering all charts and statistics to show only that specific route. Figure 7 is an example of Lot's O' Nuts to Alabama, showing 34 shipments, an average lead time of 178.7 days, a standard deviation of 1.54 days, and a delay rate of 29.4 percent on $476 in sales.

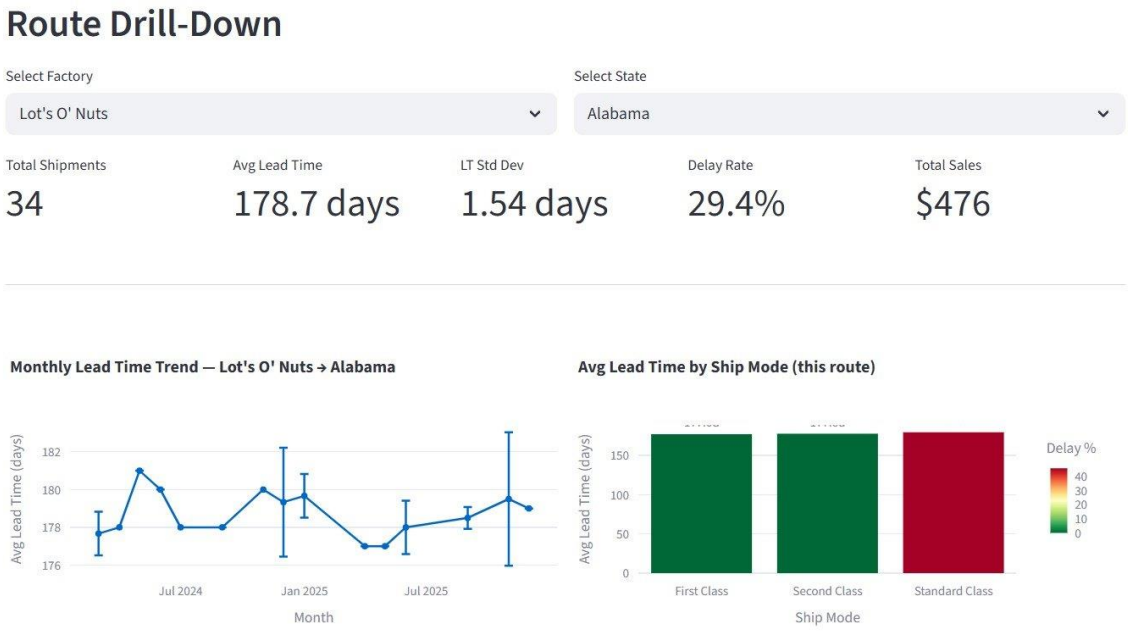

Figure 7: Tab 5: Route Drill-Down for Lot's O' Nuts → Alabama, showing monthly lead time trend and ship mode breakdown

The monthly trend chart to the left of Figure 7 indicates that lead time oscillates between 177 days and 181 days throughout 2024-2025, with a particularly high spike in mid-2024. The ship mode bar chart to the right of Figure 7 indicates that, as is true throughout the entire network as seen in Tab 4, Standard Class has a higher percentage of delays than First Class and Second Class for this particular route. An area of the display below these charts can be expanded to show all individual orders, including their lead times.

8. DISCUSSION

VIII.1 Interpretation of Findings

The narrow range of lead times between 11 days (174-185 days) implies that the distribution system has an underlying stable baseline. However, the high rate of delays at 38.8% in Standard Class, which accounts for 6,120 of the total 10,194 shipments (60%), implies that there is considerable risk in the system. Over a third of all customers in Standard Class receive their orders later than the 75th percentile benchmark.

The fact that the bottleneck is spread across all four regions implies that the delays are not caused by a particular region. Also, it is unlikely that the delays were caused by weather. It is likely that it is simply a capacity problem. Another interesting fact is that Sugar Shack has the highest delay rate of 36.4%, but it is also the factory that has the lowest volume of 33 orders.

A counterintuitive result is that the Standard Class is not the lowest cost ($4.75 average cost), despite being the slowest. Second Class is slightly more expensive ($4.83) but has substantially better performance at 7.5% vs. 38.8% delay rate.

VIII.2 Operational Recommendations

1. Migrate Standard Class to Second Class: minimal cost increase, 31.3 percentage point delay rate reduction
2. Prioritize operational review in six bottleneck states: Minnesota, Tennessee, New Jersey, Indiana, Oregon, Delaware collectively has 781 annual shipments with above-average lead times
3. Examine Sugar Shack dispatch practices: highest delay rate (36.4%) of all factories
4. Benchmark practices at The Other Factory: fastest average lead time (177.98 days), lowest delay rate (21.0%), most diverse route network (28 states)

5. Track lead time variability in addition to averages: high standard deviation routes cannot guarantee delivery date promises

VIII.3 Limitations

The modulo-365 assumes that the true lead times will fall within a single calendar year. This is probably true in the case of the candy distributor but would be false if the lead times were actually multi-year. It would have been nice to have the carrier data, the warehouse data, and the customer satisfaction data. The delay rate will automatically adjust based on the distribution of the data. This might hide the effect of gradual degradation over time.

9. CONCLUSION

In this paper, the entire design and development of the shipping route efficiency intelligence system for Nassau Candy Distributor has been presented. The entire raw CSV data containing 10,194 orders is transformed into an entirely interactive five-tab analytics system with the help of a well-structured nine-step pipeline and an efficient Streamlit dashboard.

The technical contribution was the identification and rectification of the ship date year inflation issue with the help of modulo-365 correction, allowing us to derive accurate lead time information for all orders. The entire analysis revealed that there are 196 shipping routes, six geographic bottleneck states, and that the most commonly used shipping type, Standard Class, has an impressive delay rate of 38.8%, while Same Day and First Class have a delay rate of 0%.

The entire dashboard is shown in Figures 1 to 7 and provides logistics managers with an opportunity to have an interactive analytics system that enables them to convert multi-hour analysis into seconds-long interaction. The future work would involve the integration of real-time order information and other relevant details to expand the capabilities beyond merely descriptive analytics.

10. REFERENCES

[1] McKinney, W. (2022). Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter (3rd ed.). O'Reilly Media.
[2] VanderPlas, J. (2023). Python Data Science Handbook: Essential Tools for Working with Data. O'Reilly Media.
[3] Streamlit Inc. (2024). Streamlit Documentation. Retrieved from https://docs.streamlit.io
[4] Plotly Technologies Inc. (2024). Plotly Python Open Source Graphing Library. Retrieved from https://plotly.com/python/
[5] Chopra, S., & Meindl, P. (2021). Supply Chain Management: Strategy, Planning, and Operation (7th ed.). Pearson.
[6] Harrison, A., & van Hoek, R. (2020). Logistics Management and Strategy (5th ed.). Pearson.
[7] Pandas Development Team. (2024). pandas 2.x Documentation. Retrieved from https://pandas.pydata.org/docs/